Python IP Allow-List Automation

Reproducible security automation with validation, CIDR checks, dry-run support, logging, and testing

Objective

Automate repetitive allow-list review while preserving clear change evidence. The included script validates IPv4 values, removes duplicates, checks each address against approved CIDR networks, writes an updated list, and records a JSON summary plus log.

Input entry	Result	Reason
10.10.10.15	Approved	Inside 10.10.10.0/24
10.10.10.21	Approved	Inside 10.10.10.0/24
10.10.20.5	Approved	Inside 10.10.20.0/24
192.0.2.25	Removed	Outside approved 192.0.2.0/28 range
198.51.100.10	Removed	Not in an approved network
10.10.10.15 (duplicate)	Removed	Duplicate normalized entry
bad-ip-value	Removed	Invalid IPv4 value

Generated Change Summary

Metric	Result
input_entries	7
approved_entries	3
invalid_entries_removed	1
unauthorized_entries_removed	2
duplicates_removed	1
dry_run	False

Controls Built Into the Script

- IPv4 parsing through Python's ipaddress module rather than string-pattern guesses.
- CIDR membership validation against a separate approved-network file.
- Duplicate removal to keep a deterministic allow-list.
- Dry-run support so changes can be reviewed without updating the output file.
- Structured JSON summary plus line-by-line log for invalid, duplicate, and unauthorized entries.
- A separate test verifies approved, duplicate, unauthorized, and invalid behavior.

Files

- python/ip_allowlist_manager.py
- data/sample_allowlist.txt
- data/approved_networks.txt
- outputs/updated_allowlist.txt
- outputs/allowlist_change_summary.json
- outputs/allowlist_automation.log
- tests/test_ip_allowlist_manager.py